SAMPLE PAPER (made up for testing - not a real exam)

Examination May 2025

Q.No	Question	Marks	CO	BL
1	Explain the self-attention mechanism with query, key and value vectors.	10	CO3	L2
2	What is Retrieval Augmented Generation (RAG)? Explain how document loaders, text splitters and vector stores are used in it.	10	CO4	L3
3	Explain sequential chains in LangChain with an example application.	5	CO4	L2
4	What are output parsers in LangChain? Explain the Pydantic output parser.	5	CO4	L2
5	Explain the data acquisition and text preprocessing steps of an NLP pipeline.	10	CO1	L2
6	When should you fine-tune an LLM instead of using prompt engineering? Explain with use cases.	10	CO5	L3